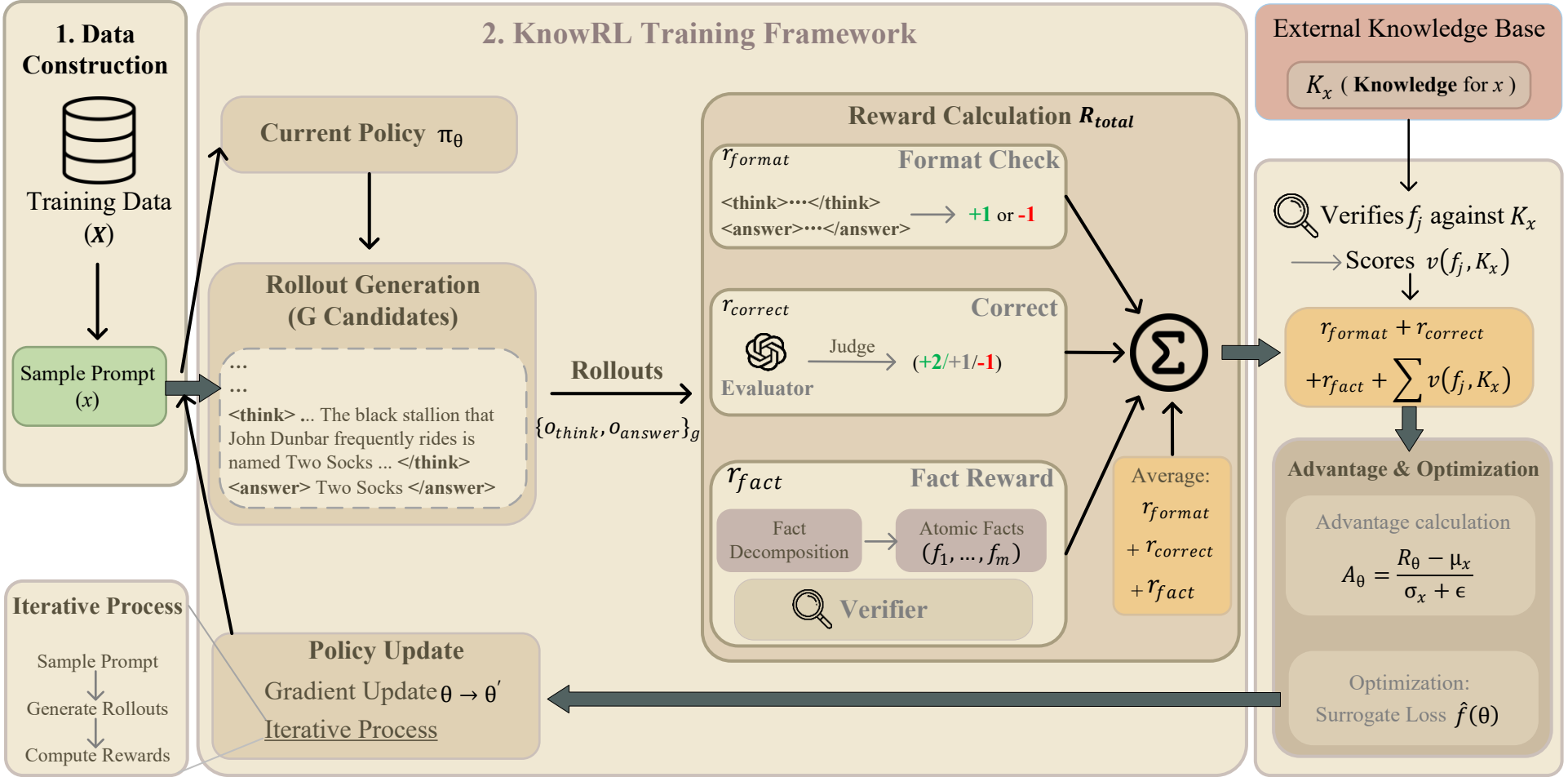

1. Data Construction
Training Data (X)
Sample Prompt (x)
2. KnowRL Training Framework
Current Policy π_θ
Rollout Generation (G Candidates)
...
...
<think> ... The black stallion that John Dunbar frequently rides is named Two Socks ... </think>
<answer> Two Socks </answer>
Rollouts
$\{o_{think}, o_{answer}\}_g$
Reward Calculation R_{total}
r_{format} Format Check
<think>····</think>
<answer>····</answer>
+1 or -1
$r_{correct}$ Correct
Judge
Evaluator
(+2/+1/-1)
r_{fact} Fact Reward
Fact Decomposition
Atomic Facts $(f_1, \ldots, f_m)$
Verifier
Average: $r_{format} + r_{correct} + r_{fact}$
External Knowledge Base
K_x (Knowledge for x)
Verifies f_j against K_x
Scores $v(f_j, K_x)$
$r_{format} + r_{correct} + r_{fact} + \sum v(f_j, K_x)$
Advantage & Optimization
Advantage calculation
$A_\theta = \frac{R_\theta - \mu_x}{\sigma_x + \epsilon}$
Optimization: Surrogate Loss $\hat{f}(\theta)$
Policy Update
Gradient Update $\theta \rightarrow \theta'$
Iterative Process
Iterative Process
Sample Prompt
Generate Rollouts
Compute Rewards